UNIFORMLY TIGHT ANCIENT SOLUTIONS OF THE NAVIER–STOKES EQUATIONS IN SELF-SIMILAR VARIABLES

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Abstract. We study bounded mild ancient solutions of the three-dimensional Navier–Stokes equations in backward self-similar variables. We establish local regularity, compactness, and stability of the Leray pressure under a uniform $L^\infty(\mathbb{R}_\tau; L^3(\mathbb{R}^3_y))$ bound. We prove that any locally smooth time-translate limit which is stationary and arises from a uniformly L^3-tight ancient solution is identically zero. The proof invokes the stationary self-similar rigidity theorem of Nečas–Růžička–Šverak. We also prove a small-data Liouville theorem for bounded mild ancient solutions, using only the mild formulation and the decay of the Ornstein–Uhlenbeck–Stokes semigroup.

1. Introduction

We consider the three-dimensional incompressible Navier–Stokes equations in backward self-similar variables. The velocity V and pressure P solve

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \qquad \operatorname{div} V = 0 \tag{1.1}$$

on $\mathbb{R}^3 \times \mathbb{R}$. This equation is the autonomous form obtained from the physical Navier–Stokes system by the change of variables

$$V(y, \tau) = \sqrt{-t}\, u(y\sqrt{-t}, t), \qquad \tau = -\log(-t), \qquad t < 0.$$

The drift term fixes the self-similar scale. The remaining scaling action is translation in the logarithmic time variable τ.

Throughout the paper repeated spatial indices are summed, and R_i denotes the i-th Riesz transform on $\mathbb{R}^3$, with Fourier symbol $i\xi_i/|\xi|$. Thus R_iR_j has symbol $-\xi_i\xi_j/|\xi|^2$, and

$$P = R_iR_j(F_{ij}) \quad \Longleftrightarrow \quad -\Delta P = \partial_i\partial_j F_{ij}$$

in distributions. All differential equations are understood in the sense of distributions unless a stronger sense is explicitly stated. Pressures are determined only modulo functions of time; a "gauge" always means the choice of such a time-dependent additive function. Constants denoted by C may change from line to line, but their displayed dependencies are the only dependencies used in the argument.

This paper does not assert a full ancient Liouville theorem for (1.1). It isolates two conditional Liouville criteria which are available under additional compactness or smallness hypotheses.

First, uniform tightness in $L^3(\mathbb{R}^3)$ permits one to pass from local smooth convergence to a global L^3 stationary self-similar profile. The theorem of Nečas, Růžička, and Šverak [9] then implies that the stationary limit is zero. Second, a bounded mild ancient solution with sufficiently small $L^\infty(\mathbb{R}^3 \times \mathbb{R})$ norm is zero. This follows from the mild formulation of (1.1) and the decay of the linear Ornstein–Uhlenbeck–Stokes semigroup.

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1.1. **Logical scope.** The results below are separated according to their hypotheses. The global source-level condition $u \in L^\infty(0,T;L^3(\mathbb{R}^3))$ is ruled out by the Escauriaza–Seregin–Šverak theorem before any ancient-profile classification is required. The small-amplitude ancient regime is ruled out by the mild formula and the damping of the centered Stokes semigroup. The stationary L^3 regime is ruled out by the stationary self-similar theorem. The uniformly L^3-tight time-dependent regime is not excluded here unless an additional argument produces a stationary limit.

The source-level facts from the preceding papers in the series are used only as inputs. The first-node paper proves pressure-normalized CKN concentration at singular points and the local Type I/Type II dichotomy [1, Theorems "Singular points carry positive local CKN concentration" and "Local blowup-rate dichotomy"]. The preceding Type I paper proves that, after the admissible pressure gauges and scale selection, a Type I singular point produces a bounded centered ancient solution with the nonvanishing normalization [2, Theorem "Type I blow-up limit and centered ancient equation" and Proposition "Nonzero ancient limit"]. Those concentration, gauge, compactness, and nontriviality conclusions are not reproved here.

1.2. **Mild formulation.** After applying the Leray projection $\mathbb{P}$, equation (1.1) becomes

$$\partial_\tau V = \Delta V - \frac{1}{2} y \cdot \nabla V - \frac{1}{2} V - \mathbb{P}\operatorname{div}(V \otimes V). \tag{1.2}$$

The linear semigroup associated with the first three terms on the right-hand side is

$$(S(a)f)(y) = e^{-a/2} \int_{\mathbb{R}^3} \frac{\exp\Big(-\frac{|z-e^{-a/2}y|^2}{4(1-e^{-a})}\Big)}{[4\pi(1-e^{-a})]^{3/2}} f(z)\,dz, \qquad a > 0. \tag{1.3}$$

The factor $e^{-a/2}$ is the damping coming from the term $-V/2$. Equivalently, $S(a) = e^{aL}$, where

$$L = \Delta - \frac{1}{2} y \cdot \nabla - \frac{1}{2}.$$

The formula shows that $S(a)$ maps bounded functions to bounded functions, preserves distributional divergence-free vector fields, and satisfies the semigroup law $S(a)S(b) = S(a+b)$. These facts follow from the change of variables representation of the Ornstein–Uhlenbeck heat kernel and the corresponding facts for the heat semigroup.

Definition 1.1 (Bounded mild ancient solution)**.** A vector field $V : \mathbb{R}^3 \times \mathbb{R} \to \mathbb{R}^3$ is a bounded mild ancient solution of (1.1) if

$$V \in L^\infty(\mathbb{R}^3 \times \mathbb{R}), \qquad \operatorname{div} V = 0$$

in distributions, and if for every $-\infty < \sigma < \tau < \infty$

$$V(\tau) = S(\tau - \sigma)V(\sigma) - \int_\sigma^\tau S(\tau - s)\mathbb{P}\operatorname{div}(V \otimes V)(s)\,ds \tag{1.4}$$

as an identity in the weak-* topology of $L^\infty(\mathbb{R}^3)$. A pressure P is then recovered locally from

$$-\Delta P = \partial_i \partial_j (V_i V_j)$$

modulo functions of time.

The weak-* formulation means that, for every $\phi \in L^1(\mathbb{R}^3;\mathbb{R}^3)$, pairing (1.4) against ϕ gives a scalar identity. The Duhamel term is interpreted by duality: for compactly supported smooth ϕ and $s < \tau$,

$$\langle S(\tau - s)\mathbb{P}\operatorname{div}(V \otimes V)(s), \phi\rangle = \langle V \otimes V(s), (S(\tau - s)\mathbb{P}\operatorname{div})^* \phi\rangle\,.$$

For fixed $\sigma < \tau$, Theorem 4.1 gives

$$\left|\langle S(\tau - s)\mathbb{P}\,\mathrm{div}(V \otimes V)(s), \phi\rangle\right| \le C\|V\|_{L^\infty}^2 e^{-(\tau-s)/2}(1 - e^{-(\tau-s)})^{-1/2}\|\phi\|_{L^1},$$

and the right-hand side is integrable for $s \in (\sigma, \tau)$. Hence the time integral in (1.4) is a well-defined weak-* integral in $L^\infty(\mathbb{R}^3)$. Once the integral has been represented by this bounded function, equality against compactly supported smooth tests extends by density of $C_c^\infty(\mathbb{R}^3)$ in $L^1(\mathbb{R}^3)$ to equality against all L^1 tests. Since $L^\infty(\mathbb{R}^3)$ is identified with the dual of $L^1(\mathbb{R}^3)$ modulo almost-everywhere equality, two bounded representatives which have the same pairing against every L^1-test agree almost everywhere. Thus no pointwise pressure representative is used in the definition of mild solution.

Remark 1.2. The mild condition fixes the whole-space pressure normalization and excludes the local pressure ambiguities which are invisible to the projected equation. For whole-space ancient limits obtained from Leray solutions, this is the Leray pressure, modulo functions of time, rather than an arbitrary local pressure with additional affine spatial components. The invariance under time-dependent pressure gauges in the source suitable-weak setting is proved in the preceding Type I paper [2, Lemma "Pressure gauges do not change the equations"].

1.3. **Main results.** The first result is a local regularity and pressure compactness statement.

Theorem 1.3 (Regularity and pressure compactness)**.** *Let V be a bounded mild ancient solution of (1.1). Then V is smooth on $\mathbb{R}^3 \times \mathbb{R}$. Moreover, if V_n is a sequence of bounded mild ancient solutions satisfying*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} < \infty$$

and, in addition,

$$\sup_n \sup_{\tau\in\mathbb{R}} \|V_n(\tau)\|_{L^3(\mathbb{R}^3)} < \infty,$$

then the Leray pressures are stable under local limits. More precisely, if $P_n = R_iR_j(V_{n,i}V_{n,j})$ are the Leray pressures and $V_n \to V$ locally uniformly, then $V \in L^\infty(\mathbb{R}_\tau; L^3(\mathbb{R}^3_y))$, its Leray pressure satisfies

$$P = R_iR_j(V_iV_j) \in L^\infty(\mathbb{R}; L^{3/2}(\mathbb{R}^3)),$$

and on every compact cylinder $K = B_R \times [\tau_1, \tau_2]$ there are functions $a_n(\tau)$ such that

$$P_n - a_n(\tau) \rightharpoonup P \quad \text{weakly in } L^q(K)$$

for every $1 < q < \infty$, after passing to a subsequence which may depend on the compact cylinder and on q. The functions a_n may be chosen measurable on $[\tau_1, \tau_2]$.

The second result records the invariance of the L^3-norm under the Navier–Stokes scaling and the self-similar change of variables.

Proposition 1.4 (Scaling of the L^3-norm)**.** *Let u be a smooth Navier–Stokes solution on $\mathbb{R}^3 \times (0, T)$, and let*

$$u_k(x, t) = \lambda_k u(x_0 + \lambda_k x, T + \lambda_k^2 t), \qquad \lambda_k \downarrow 0,$$

converge locally smoothly on $\mathbb{R}^3 \times (-\infty, 0)$ to an ancient solution U. Assume that

$$\sup_{0<t<T} \|u(t)\|_{L^3(\mathbb{R}^3)} \le A < \infty.$$

Then the self-similar representative

$$V(y, \tau) = \sqrt{-t}\,U(y\sqrt{-t}, t), \qquad \tau = -\log(-t),$$

satisfies

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \le A.$$

Theorem 1.5 (Escauriaza–Seregin–Šverak critical L^3 criterion)**.** *Let (u,p) be a finite-energy suitable weak solution of the Navier–Stokes equations on $\mathbb{R}^3\times(0,T)$. If*

$$\sup_{0<t<T} \|u(t)\|_{L^3(\mathbb{R}^3)} < \infty,$$

then no point of $\mathbb{R}^3\times\{T\}$ is singular.

Proof. This is the endpoint regularity theorem of Escauriaza–Seregin–Šverak [3]. We use it only through the logically equivalent contrapositive: if a finite-time singularity occurs at time T, then $\|u(t)\|_{L^3(\mathbb{R}^3)}$ is unbounded as $t\uparrow T$. □

Corollary 1.6 (Exclusion of the source-level critical L^3 regime)**.** *A finite-energy Type I singularity cannot occur for a suitable weak solution satisfying*

$$u\in L^\infty(0,T;L^3(\mathbb{R}^3)).$$

Consequently the analysis of Type I blow-up limits is relevant only outside the global critical class $L^\infty(0,T;L^3(\mathbb{R}^3))$.

Proof. The asserted $L^\infty(0,T;L^3(\mathbb{R}^3))$ bound is exactly the hypothesis of Theorem 1.5. Hence there is no singular point at time T, and in particular no Type I singular point. Thus the local concentration and Type I/Type II alternatives of the first-node paper are never reached under this global critical assumption [1, Theorems "Singular points carry positive local CKN concentration" and "Local blowup-rate dichotomy"]. □

Theorem 1.7 (Small bounded ancient solutions vanish)**.** *There exists an absolute constant $\varepsilon_0>0$ with the following property. If V is a bounded mild ancient solution of (1.1) and*

$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le \varepsilon_0,$$

then $V\equiv 0$.

Theorem 1.8 (Stationary limits under uniform L^3-tightness)**.** *Let V be a bounded mild ancient solution of (1.1). Assume that*

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty \tag{1.5}$$

and

$$\lim_{R\to\infty}\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3\,dy = 0. \tag{1.6}$$

Let τ_n be any sequence, and set

$$V_n(y,\tau) = V(y,\tau+\tau_n).$$

Assume that $V_n\to W$ locally smoothly on $\mathbb{R}^3\times\mathbb{R}$, and that W is independent of τ. Then $W\equiv 0$.

Corollary 1.9 (Stationary solutions under uniform L^3-tightness)**.** *Let V be a bounded mild ancient solution such that*

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty$$

and

$$\lim_{R\to\infty}\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3\,dy = 0.$$

If $V(y,\tau) = W(y)$, *then*

$$V \equiv 0.$$

Corollary 1.10 (Positive size threshold)**.** *Every nonzero bounded mild ancient solution of* (1.1) *satisfies*

$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \geq \varepsilon_0,$$

where ε_0 *is the constant in Theorem 1.7.*

Corollary 1.11 (Exclusion of the small-amplitude ancient regime)**.** *Let* V *be a bounded mild ancient solution of* (1.1)*. If*

$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \leq \varepsilon_0,$$

where ε_0 *is the constant in Theorem 1.7, then*

$$V \equiv 0.$$

Consequently no nonzero Type I ancient limit in centered variables can belong to this small-amplitude class.

Proof. The first assertion is exactly Theorem 1.7. The final sentence concerns only ancient limits obtained from a source-level Type I singular point. In that setting the nonvanishing normalization is not an extra assumption of this paper: it is the output of the scale-selection and pressure-compactness construction in the preceding Type I paper, which itself invokes the pressure-normalized CKN concentration theorem from the first-node paper [1, Theorem "Singular points carry positive local CKN concentration"] [2, Proposition "Nonzero ancient limit" and Theorem "Type I blow-up limit and centered ancient equation"]. Therefore a Type I ancient limit produced by that construction cannot both be nonzero in the cited normalization and vanish by the small-amplitude theorem. □

Corollary 1.12 (Exclusion of the stationary L^3 ancient regime)**.** *Let* V *be a bounded mild ancient solution of* (1.1)*. Suppose that* V *is stationary,* $V(y,\tau) = W(y)$*, and*

$$W \in L^3(\mathbb{R}^3).$$

Then

$$V \equiv 0.$$

Consequently no nonzero Type I ancient limit in centered variables can be both stationary and globally L^3*.*

Proof. By Theorem 1.3, V is smooth. Since it is independent of τ, $\partial_\tau V = 0$ in $\mathcal{D}'(\mathbb{R}^3\times\mathbb{R})$. By Theorem 2.3, the mild formulation implies the projected stationary equation

$$0 = \Delta W - \frac{1}{2} y \cdot \nabla W - \frac{1}{2} W - \mathbb{P}\,\mathrm{div}(W \otimes W).$$

The assumption $W \in L^3$ implies $W_i W_j \in L^{3/2}(\mathbb{R}^3)$. Since the Riesz transforms are bounded on $L^{3/2}(\mathbb{R}^3)$ [12], the Leray pressure

$$P = R_i R_j (W_i W_j)$$

belongs to $L^{3/2}(\mathbb{R}^3)$. The whole-space Helmholtz decomposition in $\mathcal{D}'(\mathbb{R}^3)$ [11, 4] gives

$$(I - \mathbb{P})\,\mathrm{div}(W \otimes W) = -\nabla P.$$

Equivalently,

$$\mathbb{P}\,\mathrm{div}(W \otimes W) = \mathrm{div}(W \otimes W) + \nabla P.$$

Substituting this identity into the projected equation gives

$$\operatorname{div}(W \otimes W) + \nabla P = \Delta W - \frac{1}{2}(W + y \cdot \nabla W)$$

in $\mathcal{D}'(\mathbb{R}^3)$. Since $\operatorname{div} W = 0$,

$$\operatorname{div}(W \otimes W)_i = \partial_j(W_i W_j) = W_j \partial_j W_i + W_i \partial_j W_j = (W \cdot \nabla) W_i.$$

Thus (6.1) holds in $\mathcal{D}'(\mathbb{R}^3)$, and Theorem 6.1 gives $W \equiv 0$. The final claim is then a direct comparison with the nonvanishing normalization supplied by the Type I blow-up limit theorem in the preceding paper, ultimately based on the positive local CKN concentration theorem at the singular point [1, Theorem "Singular points carry positive local CKN concentration"] [2, Theorem "Type I blow-up limit and centered ancient equation" and Proposition "Nonzero ancient limit"]. No additional normalization or compactness assumption is introduced in this corollary. □

2. Pressure estimates and compactness

We first record the local pressure estimates used below. The invariance of time-dependent pressure gauges for source suitable weak solutions and their parabolic rescalings is part of the preceding Type I construction [2, Lemma "Pressure gauges do not change the equations" and Lemma "Pressure gauges and local pressure bounds"]. The lemmas in this section are the whole-space Leray-pressure and smooth-ancient-profile estimates needed for the conditional arguments of the present paper.

Lemma 2.1 (Local pressure decomposition)**.** *Let V be bounded on $B_{4R} \times [\tau_1, \tau_2]$, and let $P \in L^1(B_{4R} \times [\tau_1, \tau_2])$ solve*

$$-\Delta P = \partial_i \partial_j (V_i V_j)$$

in $\mathcal{D}'(B_{4R} \times (\tau_1, \tau_2))$. Then there is a measurable function $a : (\tau_1, \tau_2) \to \mathbb{R}$ such that, for every $1 < q < \infty$ and almost every $\tau \in [\tau_1, \tau_2]$,

$$\|P(\tau) - a(\tau)\|_{L^q(B_R)} \leq C(q, R)\|V(\tau)\|^2_{L^\infty(B_{4R})} + C(q, R)\|P(\tau)\|_{L^1(B_{4R})}.$$

If P is the whole-space Leray pressure and $V \in L^\infty(\mathbb{R}_\tau; L^3(\mathbb{R}^3_y))$, then, after choosing the Leray gauge,

$$\|P - a(\tau)\|_{L^q(B_R \times [\tau_1, \tau_2])} \leq C(q, R, \tau_1, \tau_2) \left(\|V\|^2_{L^\infty(B_{4R} \times [\tau_1, \tau_2])} + \sup_{\tau \in \mathbb{R}} \|V(\tau)\|^2_{L^3(\mathbb{R}^3)} \right).$$

Here the spacetime norm is taken with the representative $(y, \tau) \mapsto P(y, \tau) - a(\tau)$.

Proof. For the first assertion it is enough to argue for almost every fixed τ, with constants independent of that τ. Indeed, testing the spacetime distributional equation against products $\zeta(\tau)\varphi(y)$ and applying Fubini shows that, for every $\varphi \in C_c^\infty(B_{4R})$, the scalar identity

$$\int_{B_{4R}} P(y, \tau)(-\Delta\varphi)(y)\, dy = \int_{B_{4R}} V_i(y, \tau) V_j(y, \tau) \partial_i \partial_j \varphi(y)\, dy$$

holds for almost every τ. Applying this first to a countable dense set of test functions in $C_c^\infty(B_{4R})$, and then using continuity of the two sides with respect to the usual test-function topology, gives a single null set outside which

$$-\Delta P(\cdot, \tau) = \partial_i \partial_j (V_i V_j)(\cdot, \tau) \quad \text{in } \mathcal{D}'(B_{4R})$$

for every test function. Choose $\chi \in C_c^\infty(B_{4R})$ with $\chi \equiv 1$ on B_{3R}, and set

$$\pi_{\text{loc}} = (-\Delta)^{-1} \partial_i \partial_j (\chi V_i V_j).$$

The inverse Laplacian is the whole-space Newtonian potential, so π_{loc} is defined as a singular integral of the compactly supported function χV_iV_j. Since V is bounded on B_{4R}, this compactly supported product belongs to $L^q(\mathbb{R}^3)$ for every $1 < q < \infty$. The Calderón–Zygmund theorem [12] gives

$$\|\pi_{\mathrm{loc}}\|_{L^q(\mathbb{R}^3)} \le C(q)\|\chi V_iV_j\|_{L^q(\mathbb{R}^3)} \le C(q,R)\|V\|^2_{L^\infty(B_{4R})}.$$

In B_{3R} we have

$$-\Delta(P - \pi_{\mathrm{loc}}) = \partial_i\partial_j(V_iV_j) - \partial_i\partial_j(\chi V_iV_j) = 0$$

in $\mathcal{D}'(B_{3R})$, because $\chi \equiv 1$ on B_{3R}. Weyl's lemma then implies that $h = P - \pi_{\mathrm{loc}}$ is harmonic in B_{3R}; this is the interior elliptic regularity implication for the Laplacian, see for instance [5]. Define

$$a(\tau) = (h)_{B_{2R}}(\tau) = |B_{2R}|^{-1}\int_{B_{2R}} h(y,\tau)\,dy.$$

The mean value property and interior estimates for harmonic functions give

$$\|h - a(\tau)\|_{L^q(B_R)} \le C(R,q)\|h - a(\tau)\|_{L^1(B_{2R})}.$$

Since $a(\tau)$ is the mean of h on B_{2R},

$$\|h - a(\tau)\|_{L^1(B_{2R})} \le \|h\|_{L^1(B_{2R})} + |B_{2R}|\,|(h)_{B_{2R}}| \le 2\|h\|_{L^1(B_{2R})}.$$

Therefore

$$\|h - a(\tau)\|_{L^q(B_R)} \le C(R,q)\|h\|_{L^1(B_{2R})}.$$

Since $h = P - \pi_{\mathrm{loc}}$, the last two estimates imply

$$\|P - a(\tau)\|_{L^q(B_R)} \le C(R,q)\|P\|_{L^1(B_{2R})} + C(R,q)\|\pi_{\mathrm{loc}}\|_{L^1(B_{2R})} + C(R,q)\|\pi_{\mathrm{loc}}\|_{L^q(B_R)}.$$

The π_{loc}-terms are bounded by the Calderón–Zygmund estimate above, and this gives the pointwise-in-time local pressure estimate. The function a is measurable because it is obtained by integrating the locally integrable function $h = P - \pi_{\mathrm{loc}}$ over B_{2R}.

Assume now that $P = R_iR_j(V_iV_j)$ is the Leray pressure and

$$M_3 = \sup_{\tau\in\mathbb{R}}\|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

We spell out the gauge because this is where the nonlocal pressure tail enters. Let K_{ij} be the homogeneous degree -3 kernel of R_iR_j, interpreted in the principal value sense. The singular-integral facts used here are the Riesz-transform Calderón–Zygmund bounds [12]. The local part $R_iR_j(\chi V_iV_j)$ is controlled exactly as above. For the exterior part we subtract a constant in x. The kernel is smooth away from the origin and satisfies, for $x \in B_R$ and $|z| > 4R$,

$$|K_{ij}(x - z) - K_{ij}(-z)| \le CR|z|^{-4}.$$

On the dyadic annulus

$$A_m = \{2^mR < |z| < 2^{m+1}R\}, \qquad m \ge 2,$$

Hölder's inequality gives

$$\int_{A_m} |V(z,\tau)|^2\,dz \le \|V(\tau)\|^2_{L^3(A_m)}|A_m|^{1/3} \le CM_3^2 2^mR.$$

Consequently

$$\sum_{m=2}^{\infty} R(2^mR)^{-4}\int_{2^mR<|z|<2^{m+1}R} |V(z,\tau)|^2\,dz \le CM_3^2R^{-2},$$

uniformly in τ. This gives a bounded far-field contribution modulo the chosen constant, with a bound depending only on M_3. Combining the near-field Calderón–Zygmund estimate and this dyadic exterior estimate gives the stated Leray-gauge bound on $B_R \times [\tau_1,\tau_2]$. □

Lemma 2.2 (Smoothing of bounded mild solutions)**.** *Let V be a bounded mild solution of* (1.1) *on $\mathbb{R}^3 \times (\tau_0, \tau_1)$. For every compact cylinder $K \Subset \mathbb{R}^3 \times (\tau_0, \tau_1)$ and every integer $m \geq 0$,*

$$\|V\|_{C^m(K)} \leq C\big(m, K, \tau_0, \tau_1, \|V\|_{L^\infty}\big).$$

The same bound holds uniformly for families with a common L^∞ bound.

Proof. Fix a compact interval $J \Subset (\tau_0, \tau_1)$ containing the time projection of K, and put $t = -e^{-\tau}$. Let $t(J) = \{-e^{-\tau} : \tau \in J\}$. On this compact negative time interval define

$$U(x,t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)).$$

The map

$$(x,t) \longmapsto (y,\tau) = \big(x/\sqrt{-t}, -\log(-t)\big)$$

is a C^∞ diffeomorphism from $\mathbb{R}^3 \times t(J)$ onto $\mathbb{R}^3 \times J$. Its spatial Jacobian at fixed time is $(-t)^{-3/2}$, and all derivatives of the map and of its inverse are bounded on compact subcylinders because $t(J) \Subset (-\infty, 0)$. We first justify that U is a bounded mild Navier–Stokes solution on $\mathbb{R}^3 \times t(J)$. If

$$x = \sqrt{-t}\, y, \qquad t = -e^{-\tau},$$

then $d\tau/dt = -1/t$ and

$$\partial_t U = (-t)^{-3/2} \left(\partial_\tau V + \frac{1}{2} V + \frac{1}{2} y \cdot \nabla_y V \right), \quad \Delta_x U = (-t)^{-3/2} \Delta_y V,$$

while

$$(U \cdot \nabla_x) U = (-t)^{-3/2} (V \cdot \nabla_y) V.$$

The pressure transforms as

$$p(x,t) = (-t)^{-1} P(x/\sqrt{-t}, -\log(-t)).$$

Consequently

$$\nabla_x p = (-t)^{-3/2} \nabla_y P, \qquad \operatorname{div}_x U = (-t)^{-1} \operatorname{div}_y V.$$

Substituting these identities in (1.1) gives

$$\partial_t U + (U \cdot \nabla) U + \nabla p = \Delta U, \qquad \operatorname{div} U = 0,$$

in $\mathcal{D}'(\mathbb{R}^3 \times t(J))$. This distributional statement is obtained by testing against compactly supported functions in (x,t), pulling the tests back by the above diffeomorphism, and using the displayed Jacobian; no boundary term occurs because the tests are compactly supported in the open time interval $t(J)$. Moreover, for $t_1 < t_2$ in $t(J)$, the corresponding logarithmic times $\tau_i = -\log(-t_i)$ satisfy $\tau_1 < \tau_2$, and testing the weak-* Duhamel formula (1.4) against the rescaled function $(-t_2)^{-3/2} \phi(\cdot/\sqrt{-t_2})$ gives, after the change of variables $x = \sqrt{-t}\, y$, the usual heat-kernel Duhamel formula

$$U(t_2) = e^{(t_2 - t_1)\Delta} U(t_1) - \int_{t_1}^{t_2} e^{(t_2 - s)\Delta} \mathbb{P} \operatorname{div}(U \otimes U)(s)\, ds$$

in weak-* $L^\infty(\mathbb{R}^3)$. The covariance of the Leray projection under translations and dilations is used here through its Fourier multiplier symbol $\delta_{ij} - \xi_i \xi_j / |\xi|^2$, which is homogeneous of degree zero. Thus U is a bounded mild Navier–Stokes solution on every compact subinterval of $t(J)$. On J, the quantities $(-t)^{1/2}$ and $(-t)^{-1/2}$ are bounded above and below by positive constants depending only on J. Hence

$$\|U\|_{L^\infty(\mathbb{R}^3 \times t(J))} \leq C(J) \|V\|_{L^\infty(\mathbb{R}^3 \times (\tau_0, \tau_1))}.$$

The interior smoothing theorem for bounded mild Navier–Stokes solutions, in the form of Kato [7] and Giga–Miyakawa [6], followed by the local parabolic bootstrap for the Stokes system

[8, 11], gives bounds for all spatial and time derivatives of U on compact subcylinders of $\mathbb{R}^3 \times t(J)$. The constants depend only on the displayed bound, the compact cylinder, and the parabolic distance to the ends of $t(J)$. The derivatives of the map $(y,\tau) \mapsto (x,t) = (y\sqrt{-t}, -e^{-\tau})$ are bounded on the chosen compact cylinder, with bounds depending only on J and the spatial size of the cylinder. Returning to (y,τ) therefore gives the asserted estimates for V. The same argument applies uniformly to any family with a common L^∞ bound. □

Lemma 2.3 (From mild to projected distributional form)**.** *Let V be a bounded mild solution of (1.1) on $\mathbb{R}^3 \times (\tau_0, \tau_1)$. If V is smooth on this cylinder, then V satisfies (1.2) in $\mathcal{D}'(\mathbb{R}^3 \times (\tau_0, \tau_1))$.*

Proof. Write

$$L = \Delta - \frac{1}{2} y \cdot \nabla - \frac{1}{2}, \qquad N(\tau) = \mathbb{P} \operatorname{div}(V \otimes V)(\tau).$$

The semigroup $S(a)$ is generated by L on test functions and, by duality, on the L^∞-weak-* side. This is the differentiation theorem for C_0-semigroups applied to compactly supported smooth tests [10]. Fix $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ and $\tau \in (\tau_0, \tau_1)$. For $h > 0$ small enough that $\tau + h < \tau_1$, the mild formula on $[\tau, \tau + h]$ gives

$$V(\tau + h) - V(\tau) = (S(h) - I)V(\tau) - \int_0^h S(h-s)N(\tau + s)\, ds.$$

Pair this identity with ϕ and divide by h. For the linear term,

$$\frac{1}{h}\langle (S(h) - I)V(\tau), \phi\rangle = \left\langle V(\tau), \frac{S(h)^*\phi - \phi}{h} \right\rangle \to \langle V(\tau), L^*\phi\rangle = \langle LV(\tau), \phi\rangle,$$

where the last equality follows by integration by parts, justified because ϕ is compactly supported and $V(\tau)$ is smooth. For the Duhamel term put $F = V \otimes V$. By definition of N,

$$\langle S(h-s)N(\tau + s), \phi\rangle = \langle F(\tau + s), (S(h-s)\mathbb{P} \operatorname{div})^*\phi\rangle.$$

For fixed $\phi \in C_c^\infty$, the Oseen-kernel representation and the Riesz-transform bounds imply

$$(S(r)\mathbb{P} \operatorname{div})^*\phi \to (\mathbb{P} \operatorname{div})^*\phi \quad \text{in } L^1(\mathbb{R}^3) \qquad (r \downarrow 0),$$

and the same family is bounded in L^1 for $0 \le r \le h_0$, with h_0 fixed. The convergence in L^1 also implies uniform tightness of this family in L^1 for small r. This is the strong continuity of the regularized Stokes kernel on smooth test data [12, 11]. To see explicitly why the limiting adjoint is integrable, use that $\mathbb{P}$ is self-adjoint on L^2 and commutes with derivatives. For matrix fields F,

$$\langle \mathbb{P} \operatorname{div} F, \phi\rangle = -\int_{\mathbb{R}^3} F_{ij}\, \partial_j(\mathbb{P}\phi)_i\, dy.$$

Hence

$$(\mathbb{P} \operatorname{div})^*\phi = -\nabla(\mathbb{P}\phi).$$

Since ϕ is compactly supported and smooth, the local part is smooth and compactly supported, while the nonlocal Riesz part has kernel derivatives decaying like $|y|^{-4}$. This decay is integrable in three dimensions, so $(\mathbb{P} \operatorname{div})^*\phi \in L^1(\mathbb{R}^3)$. Since F is bounded and smooth, $F(\tau + s) \to F(\tau)$ locally uniformly and pointwise as $s \to 0$. We spell out the limiting argument. Let $\varepsilon > 0$. By the uniform L^1-tightness of $(S(r)\mathbb{P} \operatorname{div})^*\phi$, choose R so large that the $L^1(\mathbb{R}^3 \setminus B_R)$-norm of every member of the family, and of the limit $(\mathbb{P} \operatorname{div})^*\phi$, is at most $\varepsilon/(1 + \|F\|_{L^\infty})$. On B_R, local uniform convergence of $F(\tau + s)$ and strong L^1-convergence of the adjoint kernels give convergence of the pairings. The exterior part is bounded by $C\|F\|_{L^\infty}\varepsilon$, uniformly for small s and r. Hence the integrand converges to $\langle N(\tau), \phi\rangle$, and it is bounded by an integrable constant on $0 < s < h$ for $h < h_0$. Averaging over $s \in (0,h)$ therefore preserves the limit as $h \downarrow 0$. Therefore

$$\frac{1}{h}\int_0^h \langle N(\tau+s), S(h-s)^*\phi\rangle\, ds \to \langle N(\tau),\phi\rangle.$$

The left-hand side converges to $\langle \partial_\tau V(\tau),\phi\rangle$ because V is smooth in τ. Thus

$$\langle \partial_\tau V(\tau),\phi\rangle = \langle LV(\tau),\phi\rangle - \langle N(\tau),\phi\rangle$$

for every compactly supported smooth ϕ and every $\tau \in (\tau_0,\tau_1)$. Multiplying by a scalar test function of time and integrating in τ gives (1.2) in $\mathcal{D}'(\mathbb{R}^3 \times (\tau_0,\tau_1))$. □

Lemma 2.4 (Local smooth compactness). *Let V_n be bounded mild ancient solutions of (1.1) with*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le M.$$

Then, after passing to a subsequence, V_n converges in $C^m_{\mathrm{loc}}(\mathbb{R}^3\times\mathbb{R})$ for every $m \ge 0$ to a bounded mild ancient solution V.

Proof. Theorem 2.2 gives uniform C^m-bounds on every compact cylinder, for every m. Choose an exhaustion

$$Q_\ell = B_\ell \times [-\ell,\ell], \qquad \ell = 1,2,\ldots.$$

For fixed ℓ and m, the uniform $C^{m+1}(Q_{\ell+1})$-bound gives equicontinuity of all derivatives of order at most m on Q_ℓ. Arzelà–Ascoli gives a subsequence converging in $C^m(Q_\ell)$. Extracting successively over the countable pairs (ℓ,m) and taking the diagonal subsequence yields convergence in C^m_{loc} for every m. The limit satisfies

$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le M$$

by pointwise convergence on compact subsets. It is also divergence free in the sense of distributions: for every $\varphi \in C_c^\infty(\mathbb{R}^3\times\mathbb{R})$,

$$\langle \operatorname{div} V,\varphi\rangle = -\int V\cdot\nabla\varphi = -\lim_{n\to\infty}\int V_n\cdot\nabla\varphi = 0,$$

because $\operatorname{div} V_n = 0$ distributionally.

It remains to verify that the global mild formulation is closed under this convergence. Fix $\sigma < \tau$. For every $\phi \in L^1(\mathbb{R}^3)$, local uniform convergence, the common L^∞ bound, and approximation of ϕ by compactly supported functions give

$$\int_{\mathbb{R}^3} V_n(\cdot,t)\phi\, dy \to \int_{\mathbb{R}^3} V(\cdot,t)\phi\, dy, \qquad t \in \{\sigma,\tau\}.$$

Indeed, if $\phi = \phi_R + (\phi-\phi_R)$ with $\phi_R \in C_c^\infty$ and $\|\phi-\phi_R\|_{L^1} \to 0$, then the compactly supported part converges by uniform convergence on a compact set, while the remainder is bounded by $2M\|\phi-\phi_R\|_{L^1}$. The same argument applies to $V_n\otimes V_n$ against L^1 matrix-valued test functions, with $2M^2$ in place of $2M$.

Now test (1.4) against a fixed $\phi \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$. By duality from Theorem 4.1, the adjoint of $S(a)\mathbb{P}\operatorname{div}$ satisfies

$$\|(S(a)\mathbb{P}\operatorname{div})^*\phi\|_{L^1} \le Ce^{-a/2}(1-e^{-a})^{-1/2}\|\phi\|_{L^1}.$$

For each $s \in (\sigma,\tau)$ the preceding paragraph gives

$$\langle V_n\otimes V_n(s), (S(\tau-s)\mathbb{P}\operatorname{div})^*\phi\rangle \to \langle V\otimes V(s), (S(\tau-s)\mathbb{P}\operatorname{div})^*\phi\rangle.$$

The absolute value of the integrand is bounded by

$$CM^2 e^{-(\tau-s)/2}(1-e^{-(\tau-s)})^{-1/2}\|\phi\|_{L^1}.$$

This function is integrable on (σ,τ): near $s=\tau$ it is comparable to $(\tau-s)^{-1/2}$, and away from $s=\tau$ it is bounded. Dominated convergence therefore passes the Duhamel term to the limit. The linear terms at times σ and τ pass to the limit by the first paragraph. Thus (1.4) holds for V against every compactly supported smooth test function. Since both sides are bounded functions of space, density of $C_c^\infty(\mathbb{R}^3)$ in $L^1(\mathbb{R}^3)$ extends the identity to all L^1 tests, which is exactly weak-* equality in $L^\infty(\mathbb{R}^3)$. $\square$

Proof of Theorem 1.3. Smoothness follows directly from Theorem 2.2. For each fixed τ, local uniform convergence gives

$$\int_{B_R} |V(y,\tau)|^3\,dy = \lim_{n\to\infty}\int_{B_R} |V_n(y,\tau)|^3\,dy$$

for every $R<\infty$. Letting $R\to\infty$ shows that $V\in L^\infty(\mathbb{R}_\tau;L^3(\mathbb{R}^3_y))$. More explicitly, if

$$M_3=\sup_n\sup_{\tau\in\mathbb{R}}\|V_n(\tau)\|_{L^3},$$

then monotone convergence in R yields

$$\|V(\tau)\|_{L^3}^3=\sup_{R<\infty}\int_{B_R}|V(y,\tau)|^3\,dy\le M_3^3$$

for every τ. The same smoothing lemma and the uniqueness of the locally uniform limit imply, after passing to a subsequence if necessary, that $V_n\to V$ locally smoothly. Indeed, any subsequence of $\{V_n\}$ has a further subsequence converging in C^m_{loc}, for every m, by Theorem 2.4; its C^0_{loc}-limit must be the given locally uniform limit V. Thus the subsequence used below may be chosen so that the convergence is locally smooth. Since $V_iV_j\in L^\infty(\mathbb{R};L^{3/2}(\mathbb{R}^3))$, the Calderón–Zygmund boundedness of the Riesz transforms on $L^{3/2}$ [12] defines

$$P=R_iR_j(V_iV_j)\in L^\infty(\mathbb{R};L^{3/2}(\mathbb{R}^3)).$$

We now identify the pressure limit in the Leray gauge. Fix $B_R\times I$. Let K_{ij} be the kernel of R_iR_j, and choose $\eta_R\in C^\infty(\mathbb{R}^3)$ with $\eta_R=0$ on B_{2R} and $\eta_R=1$ on $\mathbb{R}^3\setminus B_{4R}$. Set

$$b_n(\tau)=\int_{\mathbb{R}^3}K_{ij}(-z)\eta_R(z)V_{n,i}(z,\tau)V_{n,j}(z,\tau)\,dz$$

and define $b(\tau)$ analogously with V. These integrals converge absolutely and uniformly in n,τ, since on each dyadic annulus $A_\rho=\{\rho<|z|<2\rho\}$

$$\int_{A_\rho}|z|^{-3}|V_n(z,\tau)|^2\,dz\le C\rho^{-3}\|V_n(\tau)\|^2_{L^3(A_\rho)}|A_\rho|^{1/3}\le CM_3^2\rho^{-2}.$$

The integrands are measurable in τ, and the preceding summable majorant is independent of τ; hence b_n and b are measurable functions on I. Consequently the gauges

$$a_n(\tau)=b_n(\tau)-b(\tau)$$

are measurable. For $x\in B_R$, the gauged pressure is represented by

$$P_n(x,\tau)-b_n(\tau)=\mathrm{p.\,v.}\int_{\mathbb{R}^3}\big(K_{ij}(x-z)-\eta_R(z)K_{ij}(-z)\big)V_{n,i}(z,\tau)V_{n,j}(z,\tau)\,dz.$$

This is the principal-value definition of R_iR_j, with the subtracted term $b_n(\tau)$ removing only a spatially constant far-field piece; subtracting such a constant does not change the pressure gradient.

Fix $L > 4R$. Choose $\chi_L \in C_c^\infty(B_{2L})$ with $\chi_L \equiv 1$ on B_L. Split the gauged pressure into the part with factor $\chi_L(z)$ and the remaining exterior part. On the bounded set $B_{2L} \times I$, local smooth convergence implies

$$V_{n,i}V_{n,j} \to V_iV_j \quad \text{strongly in } L^q(B_{2L} \times I)$$

for every finite q. The truncated principal value operator is a Calderón–Zygmund operator applied to $\chi_L V_{n,i}V_{n,j}$, followed by restriction to B_R, plus the smooth subtraction operator with kernel $\eta_R(z)K_{ij}(-z)\chi_L(z)$. This subtraction kernel is smooth and compactly supported because χ_L is compactly supported and $\eta_R = 0$ on B_{2R}, away from the singularity of $K_{ij}(-z)$ at the origin. Calderón–Zygmund boundedness on L^q, $1 < q < \infty$, and Young's inequality for this smooth compactly supported kernel therefore show that the contribution with factor χ_L converges strongly in $L^q(B_R \times I)$ for every $1 < q < \infty$.

The exterior contribution is uniform in n. On the support of $1 - \chi_L$ one has $|z| > L$, and for $x \in B_R$

$$|K_{ij}(x - z) - K_{ij}(-z)| \le C_R|z|^{-4},$$

and therefore

$$\begin{aligned}
&\sup_{n,\tau}\sup_{x\in B_R}\left|\int_{|z|>L}[K_{ij}(x-z) - K_{ij}(-z)]V_{n,i}V_{n,j}(z,\tau)\,dz\right| \\
&\le C_R\sum_{m=0}^{\infty}(2^mL)^{-4}\int_{2^mL<|z|<2^{m+1}L}|V_n(z,\tau)|^2\,dz \\
&\le C_RM_3^2L^{-3}.
\end{aligned}$$

The same estimate holds for the limit V. Letting $n \to \infty$ for fixed L, and then $L \to \infty$, gives $P_n - b_n \to P - b$ strongly in $L^q(B_R \times I)$ for every $1 < q < \infty$. Taking $a_n = b_n - b$ gives the stated convergence to the Leray pressure P; in particular the asserted weak convergence follows. □

3. L^3 TRANSFER AND TIGHTNESS

The L^3-norm is invariant under both the Navier–Stokes parabolic scaling and the centered self-similar scaling. The parabolic scaling identities and pressure-gauge conventions for the source blow-up sequence are those of the preceding Type I paper [2, Lemma "Scaling"]; the calculation below records only the resulting global L^3 consequence used here.

Proof of Theorem 1.4. For each fixed $t < 0$, the time $T + \lambda_k^2 t$ belongs to $(0, T)$ for all sufficiently large k. For those k,

$$\|u_k(t)\|_{L^3(\mathbb{R}^3)} = \|u(T + \lambda_k^2 t)\|_{L^3(\mathbb{R}^3)} \le A.$$

The equality follows by the change of variables $X = x_0 + \lambda_k x$, for which $dX = \lambda_k^3 dx$ and $|u_k(x,t)|^3 = \lambda_k^3|u(X, T + \lambda_k^2 t)|^3$. By local smooth convergence, $u_k(\cdot,t) \to U(\cdot,t)$ uniformly on each ball B_R. Since B_R has finite measure and

$$\|u_k(\cdot,t) - U(\cdot,t)\|_{L^3(B_R)} \le |B_R|^{1/3}\|u_k(\cdot,t) - U(\cdot,t)\|_{L^\infty(B_R)},$$

the convergence is strong in $L^3(B_R)$. Hence

$$\int_{B_R}|U(x,t)|^3\,dx = \lim_{k\to\infty}\int_{B_R}|u_k(x,t)|^3\,dx \le A^3$$

for every $R < \infty$ and every $t < 0$. Since $\int_{B_R} |U(x,t)|^3\,dx$ is monotone increasing in R, the monotone convergence theorem gives $\|U(t)\|_{L^3(\mathbb{R}^3)} \le A$ for every $t < 0$. Finally,

$$\|V(\tau)\|_{L^3(\mathbb{R}^3)} = \|U(t)\|_{L^3(\mathbb{R}^3)}, \qquad t = -e^{-\tau},$$

because, with $x = y\sqrt{-t}$,

$$\int_{\mathbb{R}^3} |V(y,\tau)|^3\,dy = \int_{\mathbb{R}^3} (-t)^{3/2}|U(y\sqrt{-t},t)|^3\,dy = \int_{\mathbb{R}^3} |U(x,t)|^3\,dx.$$

□

Lemma 3.1 (Closure of uniform L^3-tightness)**.** *Let V_n be bounded mild ancient solutions of* (1.1)*. Assume that*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3_y\times\mathbb{R}_\tau)} + \sup_n \sup_{\tau\in\mathbb{R}} \|V_n(\tau)\|_{L^3(\mathbb{R}^3_y)} < \infty$$

and that the tightness in (1.6) *is uniform in n:*

$$\lim_{R\to\infty} \sup_n \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V_n(y,\tau)|^3\,dy = 0.$$

If $V_n \to V$ locally smoothly, then

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty$$

and

$$\lim_{R\to\infty} \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3\,dy = 0.$$

Proof. Let

$$M_3 = \sup_n \sup_{\tau\in\mathbb{R}} \|V_n(\tau)\|_{L^3} < \infty.$$

Fix τ. On B_R, local smooth convergence gives

$$\int_{B_R} |V(\tau)|^3\,dy = \lim_{n\to\infty} \int_{B_R} |V_n(\tau)|^3\,dy.$$

Taking the supremum over R and using monotone convergence gives

$$\|V(\tau)\|_{L^3}^3 = \sup_{R<\infty} \int_{B_R} |V(\tau)|^3\,dy \le M_3^3.$$

Thus V has the same uniform L^3 bound. For tightness, choose R large so that

$$\sup_n \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V_n(y,\tau)|^3\,dy \le \varepsilon.$$

Local smooth convergence gives pointwise convergence everywhere after fixing τ. Fatou's lemma on the exterior domain gives

$$\int_{|y|>R} |V(\tau)|^3\,dy \le \liminf_{n\to\infty} \int_{|y|>R} |V_n(\tau)|^3\,dy \le \varepsilon.$$

The choice of R was made only from the uniform tail bound for the sequence $\{V_n\}$, and therefore is independent of the fixed time τ. Hence

$$\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3\,dy \le \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, the displayed uniform tail limit is zero. □

4. The small ancient Liouville theorem

The proof of Theorem 1.7 uses only the mild formulation and the linear decay of the centered Stokes flow. No spatial tightness is needed.

Lemma 4.1 (Centered Stokes estimates). *There is an absolute constant C_0 such that for every $a > 0$,*

$$\|S(a)f\|_{L^\infty} \le e^{-a/2}\|f\|_{L^\infty},$$

and

$$\|S(a)\mathbb{P}\operatorname{div} F\|_{L^\infty} \le C_0 e^{-a/2}(1-e^{-a})^{-1/2}\|F\|_{L^\infty}. \tag{4.1}$$

Consequently

$$C_* := C_0\int_0^\infty e^{-a/2}(1-e^{-a})^{-1/2}\,da < \infty. \tag{4.2}$$

Proof. The first estimate follows immediately from the positive kernel formula (1.3), since the Gaussian factor has total mass one. The Oseen-kernel estimates used below are the classical heat-kernel bounds for the Stokes semigroup; see, for example, [11, 4].

For the second estimate, first take $F \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^{3\times 3})$. The scalar Ornstein–Uhlenbeck semigroup in (1.3) may be written as

$$S(a)g(y) = e^{-a/2}\big(e^{(1-e^{-a})\Delta}g\big)(e^{-a/2}y).$$

The Leray projection is the Fourier multiplier

$$(\widehat{\mathbb{P}f})_i(\xi) = \left(\delta_{ij} - \frac{\xi_i\xi_j}{|\xi|^2}\right)\widehat{f}_j(\xi), \qquad \xi \ne 0.$$

It is homogeneous of degree zero and therefore commutes with the heat multiplier $e^{-s|\xi|^2}$ and with the derivative multipliers $i\xi_k$. Therefore, for $s = 1-e^{-a}$,

$$S(a)\mathbb{P}\operatorname{div} F(y) = e^{-a/2}\big(e^{s\Delta}\mathbb{P}\operatorname{div} F\big)(e^{-a/2}y).$$

The operator $e^{s\Delta}\mathbb{P}\operatorname{div}$ has the Oseen derivative kernel

$$K_s(x) = s^{-2}K_1(x/\sqrt{s}),$$

with

$$\|K_s\|_{L^1(\mathbb{R}^3)} \le Cs^{-1/2}.$$

This scaling follows from the single derivative in div: the heat kernel contributes $s^{-3/2}$, the derivative contributes $s^{-1/2}$, and the change of variables in the L^1-norm contributes $s^{3/2}$. Young's inequality gives

$$\|e^{s\Delta}\mathbb{P}\operatorname{div} F\|_{L^\infty} \le \|K_s\|_{L^1}\|F\|_{L^\infty} \le Cs^{-1/2}\|F\|_{L^\infty}.$$

The dilation $y \mapsto e^{-a/2}y$ has L^∞-operator norm one, and the amplitude factor in $S(a)$ gives $e^{-a/2}$. Substituting $s = 1-e^{-a}$ proves (4.1) for smooth compactly supported F. Since $K_s \in L^1(\mathbb{R}^3)$, the convolution formula defines $e^{s\Delta}\mathbb{P}\operatorname{div} F$ pointwise almost everywhere for every $F \in L^\infty(\mathbb{R}^3;\mathbb{R}^{3\times 3})$, and Young's inequality applies directly to that convolution. Therefore the same estimate holds for all bounded matrix fields F.

Finally,

$$1-e^{-a} \sim a \quad (a \downarrow 0), \qquad 1-e^{-a} \to 1 \quad (a\to\infty).$$

Thus the integrand in (4.2) is comparable to $a^{-1/2}$ near 0 and to $e^{-a/2}$ for large a, and the integral is finite. □

Proof of Theorem 1.7. Let

$$M = \|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}.$$

Apply the mild formula (1.4) on $[\tau - T, \tau]$:

$$V(\tau) = S(T)V(\tau - T) - \int_0^T S(a)\mathbb{P}\,\mathrm{div}(V\otimes V)(\tau - a)\,da.$$

The identity is an equality in weak-* L^∞, and the right-hand side is represented by an L^∞ function because of Theorem 4.1. Therefore the difference between $V(\tau)$ and the displayed right-hand side pairs to zero against every $L^1(\mathbb{R}^3)$ test, hence the two bounded representatives are equal almost everywhere. It is consequently legitimate to estimate the L^∞-norm of $V(\tau)$ by the L^∞-norm of that representative. The first term is bounded by $e^{-T/2}M$. For the Duhamel term, Theorem 4.1 and $\|V\otimes V\|_{L^\infty} \le M^2$ give

$$\|V(\tau)\|_{L^\infty} \le e^{-T/2}M + C_0M^2\int_0^T e^{-a/2}(1-e^{-a})^{-1/2}\,da.$$

Taking the supremum in τ gives

$$M \le e^{-T/2}M + C_*M^2.$$

The solution is ancient, so T is arbitrary. Letting $T\to\infty$ gives

$$M \le C_*M^2.$$

Choose $\varepsilon_0 = (2C_*)^{-1}$. If $0 < M \le \varepsilon_0$, then $1 \le C_*M \le 1/2$, a contradiction. Hence $M = 0$, and $V \equiv 0$. $\square$

5. Gaussian identity under rapid decay

The following identity is not needed for Theorem 1.7, but it is the weighted energy identity naturally associated with the Gaussian measure in self-similar variables.

Proposition 5.1 (Gaussian energy identity)**.** *Let V be a smooth bounded mild ancient solution of (1.1), with pressure P. Assume that, for every compact interval $I\subset\mathbb{R}$ and every integer $m\ge 0$,*

$$\sup_{\tau\in I}\sup_{y\in\mathbb{R}^3}(1+|y|)^m\left(|\nabla^m V(y,\tau)| + |\nabla^m P(y,\tau)|\right) < \infty.$$

Set

$$G(y) = e^{-|y|^2/4}.$$

Then

$$\frac{d}{d\tau}\int_{\mathbb{R}^3}|V(y,\tau)|^2G(y)\,dy = -2\int_{\mathbb{R}^3}|\nabla V|^2G\,dy - \int_{\mathbb{R}^3}|V|^2G\,dy - \frac{1}{2}\int_{\mathbb{R}^3}(|V|^2+2P)(V\cdot y)G\,dy. \tag{5.1}$$

Proof. All integrations by parts below may first be performed on B_R with a cutoff $\chi_R\in C_c^\infty(B_{2R})$, $\chi_R = 1$ on B_R, $\chi_R = 0$ outside B_{2R}, and $|\nabla^k\chi_R| \le C_kR^{-k}$. The assumed polynomial decay of all derivatives, together with the Gaussian factor, makes every boundary and cutoff-error term tend to zero as $R\to\infty$. Thus the formal whole-space integrations by parts are justified.

Multiply (1.1) by $2VG$ and integrate over $\mathbb{R}^3$. Smoothness in τ and dominated convergence give

$$2\int_{\mathbb{R}^3}\partial_\tau V\cdot VG\,dy = \frac{d}{d\tau}\int_{\mathbb{R}^3}|V|^2G\,dy.$$

Since $\operatorname{div} V = 0$,

$$2\int (V\cdot\nabla)V\cdot VG = \int V\cdot\nabla(|V|^2)G = -\int |V|^2 V\cdot\nabla G.$$

The pressure term gives

$$2\int \nabla P\cdot VG = -2\int PV\cdot\nabla G.$$

For the diffusion term,

$$\begin{aligned}2\int \Delta V\cdot VG &= -2\int |\nabla V|^2 G - 2\int \partial_k V_i V_i \partial_k G\\ &= -2\int |\nabla V|^2 G - \int \partial_k(|V|^2)\partial_k G\\ &= -2\int |\nabla V|^2 G + \int |V|^2\Delta G.\end{aligned}$$

Finally,

$$\begin{aligned}-\int (V + y\cdot\nabla V)\cdot VG &= -\int |V|^2 G - \frac{1}{2}\int y\cdot\nabla(|V|^2)G\\ &= -\int |V|^2 G + \frac{1}{2}\int |V|^2(3G + y\cdot\nabla G).\end{aligned}$$

Using

$$\nabla G = -\frac{y}{2}G, \qquad \Delta G = \left(\frac{|y|^2}{4} - \frac{3}{2}\right)G,$$

the pure quadratic terms are

$$\int |V|^2\Delta G - \int |V|^2 G + \frac{1}{2}\int |V|^2(3G + y\cdot\nabla G) = -\int |V|^2 G.$$

The convection and pressure terms are

$$-\int |V|^2 V\cdot\nabla G - 2\int PV\cdot\nabla G = \frac{1}{2}\int (|V|^2 + 2P)(V\cdot y)G.$$

Moving the non-time-derivative terms to the right-hand side of the equation gives (5.1). □

6. Stationary self-similar rigidity

We now recall the stationary theorem of Nečas, Růžička, and Šverak [9].

Theorem 6.1 (Stationary self-similar rigidity). *Let $W \in L^3(\mathbb{R}^3)$ be divergence free. Assume that there exists $P \in L^{3/2}_{\mathrm{loc}}(\mathbb{R}^3)$ such that*

$$(W\cdot\nabla)W + \nabla P = \Delta W - \frac{1}{2}(W + y\cdot\nabla W) \tag{6.1}$$

in $\mathcal{D}'(\mathbb{R}^3)$. Then

$$W \equiv 0.$$

Remark 6.2. The hypothesis $W \in L^3(\mathbb{R}^3)$ is essential for this direct invocation. Without a global integrability condition, Theorem 6.1 does not apply directly.

Proof of Theorem 1.8. First note that each time translate $V_n(y,\tau) = V(y,\tau+\tau_n)$ is again a bounded mild ancient solution. The L^∞ bound is unchanged, and $\operatorname{div} V_n = 0$ in distributions because translation in τ does not affect the spatial divergence. If $\sigma < \tau$, applying (1.4) to V on the interval $[\sigma+\tau_n, \tau+\tau_n]$ gives

$$V_n(\tau) = S(\tau-\sigma)V_n(\sigma) - \int_\sigma^\tau S(\tau-s)\mathbb{P}\operatorname{div}(V_n \otimes V_n)(s)\,ds$$

after the change of variables $s \mapsto s+\tau_n$ in the weak-* time integral. Thus V_n satisfies the same mild formulation as V.

The assumptions (1.5) and (1.6) are invariant under translations in τ. Hence the sequence V_n has the same L^∞, uniform L^3, and uniform tightness bounds as V. By Theorem 3.1, the local smooth limit W satisfies the same global L^3 bound. Since W is independent of τ, write $W(y,\tau) = W_0(y)$. Then

$$W_0 \in L^3(\mathbb{R}^3).$$

We verify that this stationary function satisfies the stationary self-similar equation in the distributional sense required by Theorem 6.1. Since W is a locally smooth limit of the bounded mild solutions V_n, the limit-passage argument in Theorem 2.4 applies to this already convergent sequence and shows that W is itself a bounded mild ancient solution. By Theorem 2.3, W satisfies the projected distributional equation

$$\partial_\tau W = \Delta W - \frac{1}{2}y\cdot\nabla W - \frac{1}{2}W - \mathbb{P}\operatorname{div}(W\otimes W).$$

Because $W(y,\tau) = W_0(y)$, its time derivative vanishes in $\mathcal{D}'(\mathbb{R}^3\times\mathbb{R})$: for $\Phi \in C_c^\infty(\mathbb{R}^3\times\mathbb{R};\mathbb{R}^3)$,

$$\langle \partial_\tau W, \Phi\rangle = -\int_{\mathbb{R}}\int_{\mathbb{R}^3} W_0(y)\cdot\partial_\tau\Phi(y,\tau)\,dy\,d\tau = -\int_{\mathbb{R}^3} W_0(y)\cdot\left(\int_{\mathbb{R}}\partial_\tau\Phi(y,\tau)\,d\tau\right)dy = 0.$$

For the stationary velocity $W(y,\tau) = W_0(y)$, the Leray pressure

$$P_0 = R_iR_j(W_{0,i}W_{0,j})$$

belongs to $L^{3/2}(\mathbb{R}^3)$, because $W_0 \in L^3$, hence $W_{0,i}W_{0,j} \in L^{3/2}(\mathbb{R}^3)$, and the Riesz transforms are bounded on $L^{3/2}$ [12]. In $\mathcal{D}'(\mathbb{R}^3)$,

$$(I-\mathbb{P})\operatorname{div}(W_0\otimes W_0) = -\nabla P_0, \qquad \mathbb{P}\nabla P_0 = 0.$$

This is the Helmholtz decomposition in the whole space [11, 4]. Hence

$$\mathbb{P}\operatorname{div}(W_0\otimes W_0) = \operatorname{div}(W_0\otimes W_0) + \nabla P_0.$$

Substituting this identity into the projected stationary equation converts it into

$$\operatorname{div}(W_0\otimes W_0) + \nabla P_0 = \Delta W_0 - \frac{1}{2}(W_0 + y\cdot\nabla W_0)$$

in $\mathcal{D}'(\mathbb{R}^3)$.

For every test function $\varphi \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$, testing the resulting distributional identity against $\psi(\tau)\varphi(y)$, where $\psi \in C_c^\infty(\mathbb{R})$ and $\int \psi\,d\tau = 1$, leaves the same spatial weak form because all terms in the identity are independent of τ. Finally, $\operatorname{div} W_0 = 0$ gives, pointwise because W_0 is smooth,

$$\operatorname{div}(W_0\otimes W_0)_i = \partial_j(W_{0,i}W_{0,j}) = W_{0,j}\partial_j W_{0,i} + W_{0,i}\partial_j W_{0,j} = (W_0\cdot\nabla)W_{0,i}$$

and hence the same identity holds in $\mathcal{D}'(\mathbb{R}^3)$. Therefore

$$(W_0\cdot\nabla)W_0 + \nabla P_0 = \Delta W_0 - \frac{1}{2}(W_0 + y\cdot\nabla W_0), \qquad \operatorname{div} W_0 = 0.$$

Thus W_0 is precisely a stationary self-similar profile in the class covered by Theorem 6.1. Consequently $W_0 \equiv 0$, and hence $W \equiv 0$. ☐

Proof of Theorem 1.9. Apply Theorem 1.8 with $\tau_n \equiv 0$. ☐

Proof of Theorem 1.10. This is the contrapositive of Theorem 1.7. ☐

7. Concluding remarks

These results do not by themselves imply a Type I exclusion theorem. They prove two conditional Liouville statements for bounded mild ancient solutions of (1.1). A stationary local limit is zero whenever uniform L^3-tightness supplies the global L^3 hypothesis in the Nečas–Růžička–Šverak theorem. Independently, a mild ancient solution with sufficiently small $L^\infty(\mathbb{R}^3 \times \mathbb{R})$ norm is zero. Thus any nontrivial bounded mild ancient solution must carry a definite L^∞-size, and any argument which produces a stationary limit must also provide enough spatial compactness to retain the critical L^3 bound.

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